

AWS - 1

1. Amazon Kinesis Data Firehose can only write to Amazon S3, Amazon Redshift, Amazon Elasticsearch or Splunk.
2. You can't have applications consuming data streams from Amazon Kinesis Data Firehose, that's the job of Amazon Kinesis Data Streams.
3. KDS - By default, records of a stream are accessible for up to 24 hours from the time they are added to the stream (can be raised to 7 days by enabling extended data retention).
4. The following URL paths can be used to get the instance meta data and user data from within the instance:
<http://169.254.169.254/latest/meta-data/> <http://169.254.169.254/latest/user-data/>
Further, you can get the instance public IP via the URL - <http://169.254.169.254/latest/meta-data/public-ipv4>
5. Amazon S3 supports the following destinations where it can publish events:
Amazon Simple Notification Service (Amazon SNS) topic Amazon Simple Queue Service (Amazon SQS) queue AWS Lambda
Currently, the Standard Amazon SQS queue is only allowed as an Amazon S3 event notification destination, whereas the FIFO SQS queue is not allowed.
6. Application Load Balancer can automatically distribute incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions.
7. Setup an Amazon CloudWatch alarm to monitor the health status of the instance. In case of an Instance Health Check failure, an EC2 Reboot CloudWatch Alarm Action can be used to reboot the instance
8. Using Amazon CloudWatch alarm actions, you can create alarms that automatically stop, terminate, reboot, or recover your Amazon EC2 instances.
9. Use Amazon Transcribe to convert audio files to text and Amazon Athena to understand the underlying customer sentiments.
10. Amazon Transcribe is an automatic speech recognition (ASR) service. One key feature of the service is called speaker identification, which you can use to label each individual speaker when transcribing multi-speaker audio files. You can specify Amazon Transcribe to identify 2–10 speakers in the audio clip.
11. Amazon Alexa cannot be used as an Automatic Speech Recognition (ASR) service, though Alexa internally uses ASR for its working.
12. AWS Snowball Edge Storage Optimized is the optimal choice if you need to securely and quickly transfer dozens of terabytes to petabytes of data to AWS. It provides up to 80 TB of usable HDD storage, 40 vCPUs, 1 TB of SATA SSD storage, and up to 40 Gb network connectivity to address large scale data transfer.
13. Amazon S3 delivers strong read-after-write consistency automatically.
14. A process replaces an existing object and immediately tries to read it. Amazon S3 always returns the latest version of the object.
15. An Amazon S3 Glacier vault is a container for storing archives.
16. Amazon S3 Glacier Vault Lock allows you to easily deploy and enforce compliance controls for individual Amazon S3 Glacier vaults with a vault lock policy. You can specify controls such as "write once read many" (WORM) in a vault lock policy and lock the policy from future edits.
17. Amazon S3 Glacier to deliver 99.999999999% durability.
18. Amazon S3 access control lists (ACLs) enable you to manage access to buckets and objects.
19. You can quickly create clones of an Aurora DB by using the database cloning feature. Cloning is much faster than a manual snapshot of the DB cluster. You cannot clone databases across AWS regions. The clone databases must be created in the same region as the source databases. Currently, you are limited to 15 clones based on a copy, including clones based on other clones. After that, only copies can be created. However, each copy can also have up to 15 clones.
20. Using Backtracking, you can "rewind" the DB cluster to any time you specify. One of the major advantages of backtracking is that it can rewind the DB cluster much faster compared to restoring a DB cluster via point-in-time restore (PITR) or via a manual DB cluster snapshot, which can take hours. Backtracking a DB cluster doesn't require a new DB cluster and rewinds the DB cluster in minutes.
21. If the Auto Scaling group (ASG) is using EC2 as the health check type and the Application Load Balancer (ALB) is using its in-built health check, there may be a situation where the ALB health check fails because the health check pings fail to receive a response from the instance. At the same time, ASG health check can come back as successful because it is based on EC2 based health check. Therefore, in this scenario, the ALB will remove the instance from its inventory, however, the Auto Scaling Group will fail to provide the replacement instance. It is recommended to use ALB based health checks for both Auto Scaling group and Application Load Balancer. If both the Auto Scaling group and Application Load Balancer use ALB based health checks.
22. Application Load Balancer cannot use EC2 based health checks
23. Amazon EC2 instance store for maximum performance -- Some instance types use NVMe or SATA-based solid-state drives (SSD) to deliver high random I/O performance. It is block-level storage volume.
24. When a host needs to send many records per second (RPS) to Amazon Kinesis, simply calling the basic PutRecord API action in a loop is inadequate. To reduce overhead and increase throughput, the application must batch records and implement parallel HTTP requests. This will increase the efficiency overall and ensure you are optimally using the shards.
25. placement strategies:
A Spread placement group is a group of instances that are each placed on distinct racks, with each rack having its own network and power source.
A spread placement group can span multiple Availability Zones in the same Region. You can have a maximum of seven running instances per Availability Zone per group. Therefore, to deploy 15 Amazon EC2 instances in a single Spread placement group, the company needs to use 3 Availability Zones.
26. For a small monthly monitoring and automation fee per object, Amazon S3 monitors access patterns of the objects in Amazon S3 Intelligent-Tiering and moves the ones that have not been accessed for 30 consecutive days to the infrequent access tier. If an object in the infrequent access tier is accessed, it is automatically moved back to the frequent access tier. There are no retrieval fees when using the Amazon S3 Intelligent-Tiering storage class, and no additional tiering fees when objects are moved between access tiers. It is the ideal storage class for long-lived data with access patterns that are unknown or unpredictable.
27. Use AWS WAF geo match statement listing the countries that you want to block
28. Use AWS WAF IP set statement that specifies the IP addresses that you want to allow through.
29. A NAT gateway has the following characteristics and limitations:
A NAT gateway supports 5 Gbps of bandwidth and automatically scales up to 45 Gbps.
You can associate exactly one Elastic IP address with a NAT gateway.
A NAT gateway supports the following protocols: TCP, UDP, and ICMP.
You cannot associate a security group with a NAT gateway.
You can use a network access control list (network ACL) to control the traffic to and from the subnet in which the NAT gateway is located.
A NAT gateway can support up to 55,000 simultaneous connections to each unique destination.
30. Most important point:- By combining the results of health checks of your Amazon EC2 instances and your ELBs, Amazon

- Route 53 DNS Failover can evaluate the health of the load balancer and the health of the application running on the Amazon EC2 instances behind it.
31. AWS Trusted Advisor recommends certain configuration changes by comparing your system configurations to AWS Best practices
 32. Depending on your Region, your Amazon S3 website endpoints follow one of these two formats.
s3-website dash (-) Region - <http://bucket-name.s3-website.Region.amazonaws.com>
s3-website dot (.) Region - <http://bucket-name.s3-website-Region.amazonaws.com>
 33. Amazon S3 Glacier Deep Archive is up to 75% less expensive than Amazon S3 Glacier and provides retrieval within 12 hours using the Standard retrieval speed. You may also reduce retrieval costs by selecting Bulk retrieval, which will return data within 48 hours.
 34. Use an active-passive failover configuration when you want a primary resource or group of resources to be available the majority of the time and you want a secondary resource or group of resources to be on standby in case all the primary resources become unavailable. When responding to queries, Amazon Route 53 includes only healthy primary resources. If all the primary resources are unhealthy, Route 53 begins to include only the healthy secondary resources in response to DNS queries.
 35. Non-root EBS volumes remain available even after you terminate an instance to which the volumes were attached.
 36. AWS owned keys are not stored in your AWS account. They are part of a collection of KMS keys that AWS owns and manages for use in multiple AWS accounts. AWS services can use AWS owned keys to protect your data. AWS owned keys used by DynamoDB are rotated every year (approximately 365 days).
 37. Redis, which stands for Remote Dictionary Server, is a fast, open-source, in-memory key-value data store for use as a database, cache, message broker, and queue. Memcached, a high-performance distributed memory cache service.
 38. By default, cross-zone load balancing is enabled for Application Load Balancer and disabled for Network Load Balancer
 39. A launch configuration is an instance configuration template that an Auto Scaling group uses to launch Amazon EC2 instances. When you create a launch configuration, you specify information for the instances. Include the ID of the Amazon Machine Image (AMI), the instance type, a key pair, one or more security groups, and a block device mapping. It is not possible to modify a launch configuration once it is created. The correct option is to create a new launch configuration to use the correct instance type. Then modify the Auto Scaling group to use this new launch configuration. Lastly to clean-up, just delete the old launch configuration as it is no longer needed.
 40. Kinesis is better for streaming data since queues aren't meant for real-time streaming of data.
 41. Use cross-Region Read Replicas
in addition to using Read Replicas to reduce the load on your source database instance, you can also use Read Replicas to implement a DR solution for your production DB environment. If the source DB instance fails, you can promote your Read Replica to a standalone source server. Read Replicas can also be created in a different Region than the source database. Using a cross-Region Read Replica can help ensure that you get back up and running if you experience a regional availability issue.
 42. The automated backup feature of Amazon RDS enables point-in-time recovery for your database instance. Amazon RDS will back up your database and transaction logs and store both for a user-specified retention period. If it's a Multi-AZ configuration, backups occur on standby to reduce the I/O impact on the primary. Amazon RDS supports Cross-Region Automated Backups. Manual snapshots and Read Replicas are also supported across multiple Regions.
 43. Database cloning is only available for Amazon Aurora and not for Amazon RDS.
 44. Network Load Balancer is best suited for use-cases involving low latency and high throughput workloads that involve scaling to millions of requests per second. Network Load Balancer operates at the connection level (Layer 4), routing connections to targets - Amazon EC2 instances, microservices, and containers - within Amazon Virtual Private Cloud (Amazon VPC) based on IP protocol data.
 45. Both Amazon ElastiCache for Redis and Amazon ElastiCache for Memcached are HIPAA Eligible.
 46. Amazon EBS volumes are Availability Zone (AZ) locked
 47. An IAM user with full administrator access can perform almost all AWS tasks except a few tasks designated only for the root account user. Some of the AWS tasks that only a root account user can do are as follows: change account name or root password or root email address, change AWS support plan, close AWS account, enable AWS Multi-Factor Authentication (AWS MFA) on S3 bucket delete, create Cloudfront key pair, register for GovCloud. Even though the DevOps engineer is part of the admin group, he cannot configure an Amazon S3 bucket to enable AWS MFA delete or close the company's AWS account.
 48. AWS Trusted Advisor inspects your AWS environment and makes recommendations for saving money, improving system performance, or closing security gaps. It scans your AWS infrastructure and compares it to AWS Best practices in five categories (Cost Optimization, Performance, Security, Fault Tolerance, Service limits) and then provides recommendations.
 49. You cannot set up multiple consumers for Amazon Kinesis Data Firehose delivery streams as it can dump data in a single data repository at a time,
 50. Amazon Elastic Block Store (Amazon EBS) provides block-level storage volumes for use with Amazon EC2 instances. When you create an encrypted Amazon EBS volume and attach it to a supported instance type, data stored at rest on the volume, data moving between the volume and the instance, snapshots created from the volume and volumes created from those snapshots are all encrypted. It uses AWS Key Management Service (AWS KMS) customer master keys (CMK) when creating encrypted volumes and snapshots. Encryption operations occur on the servers that host Amazon EC2 instances, ensuring the security of both data-at-rest and data-in-transit between an instance and its attached Amazon EBS storage.
 51. DynamoDB can handle more than 10 trillion requests per day and can support peaks of more than 20 million requests per second.
 52. Amazon DynamoDB is horizontally scalable, has a DynamoDB streams capability and is multi-AZ by default. On top of it, we can adjust the RCU and WCU automatically using Auto Scaling.
 53. Amazon Aurora features a distributed, fault-tolerant, self-healing storage system that auto-scales up to 64TB per database instance.
 54. If the writer instance in a cluster becomes unavailable, Aurora automatically promotes one of the reader instances to take its place as the new writer.
 55. There are no standby instances in Aurora. Aurora performs an automatic failover to a read replica when a problem is detected.
 56. Metadata, which can be included with the object, is not encrypted while being stored on Amazon S3.
 57. Amazon S3 can encrypt data in transit using HTTPS (TLS)
 58. When failing over, Amazon RDS simply flips the canonical name record (CNAME) for your DB instance to point at the standby, which is in turn promoted to become the new primary. Multi-AZ means the URL is the same, the failover is automated, and the CNAME will automatically be updated to point to the standby database.